R13

Code No: 113BS

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, April/May - 2018
DIGITAL LOGIC DESIGN
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Find the possible base so that the expression $(292)_{10}=(204)_b$. [2]
b) Construct AND, OR and NOT gate using universal gates. [3]
c) Implement Ex-OR with NAND gates. [2]
d) List the max terms of the function F(x,y,z)=xyz+x′z. [3]
e) Define combinational circuits and give examples. [2]
f) Design a half adder circuit. [3]
g) Differentiate between combinational and sequential circuits. [2]
h) What are direct inputs in a flip-flop and why they are used? [3]
i) What is the difference between PLA and PAL? [2]
j) Design a circuit using ROM that accepts a 2-bit number and generates a binary number equal to the square of the input number. [3]

PART-B

(50 Marks)

2.a) Represent $(8620)_{10}$ in BCD and Excess-3 code.
b) With an example show how Hamming Code can be used to correct single error and detect double error. [6+4]

OR

3.a) Convert $(FACE07C2)_H$ to binary and Octal.
b) Find the complement of F=x′yz′+ x′y′z and realize the function using gates.
c) Construct a 7-bit single error correcting code to represent the decimal digits by using Excess-3 code words. Use even parity check. [2+4+4]

4.a) Implement the function F(w,x,y,z)=∑(0, 2, 5, 7, 10, 13) using two level forms of NAND-AND and NOR-OR.
b) Simplify F(w,x,y,z)=∑(0,2,5, 6, 7, 8, 10) in POS. [5+5]

OR

5.a) Find all the prime implicants for the Boolean function.
F(w,x,y,z)=∑(0,2,4,5,6,7,8,10,13,15) and find which are essential.
b) Simplify BD+BCD′+AB′C′D′ and implement using two level NAND gates. [5+5]

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6.a) What is Multiplexer? Implement a full subtractor with two 4×1 multiplexers.
b) Construct a 12-bit Magnitude comparator using 4-bit magnitude comparators. [5+5]

OR

7.a) Construct 4×16 line decoder using 3×8 line decoders.
b) Design a BCD to Excess-3 code convertor. [5+5]

8. Design a circuit using JK flip-flops which has one input x and generates a high output if the sequence 0101 is detected. [10]

OR

9.a) Design a 4-bit Ring Counter.
b) Design a T flip-flop using JK flip-flop. Use k-maps for the design. [5+5]

10. Realize the functions using PAL [10]
$F_1(A,B,C, D)= \Sigma(2, 12, 13)$
$F_2(A,B,C,D)= \Sigma(7, 8, 9, 10, 11, 12, 13, 14, 15)$
$F_3(A,B,C,D)= \Sigma(0,2, 3, 4, 5, 6, 7, 8, 10, 11, 15)$
$F_4(A,B,C)= \Sigma(1,2,8,12,13)$

OR

11. List the PLA programming table for the BCD to Excess-3 converter. [10]

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